

Evolution Cheats Imagination: Auditing Safety Gaps in a Learned Walker2d Model

Anonymous Author(s)

ABSTRACT

Evolutionary search in a learned world model can select policies whose imagined safety does not survive execution. We audit this effect in Walker2d by evolving residual policies inside a frozen visual latent model and replaying selected policies from identical saved simulator states. A prospectively locked study crosses latent noise and a return-range safety penalty over 20 paired search seeds, three population sizes, and 320 fresh final source episodes. Increasing search from population 16 at generation 1 to population 256 at generation 16 increases the control’s real-minus-imagined failure gap by 10.39 percentage points (adjusted 97.5% interval: 6.70 to 14.25). Noise plus penalty reduces this amplification by 7.22 points (2.49 to 12.18). At high search pressure, actual failure falls from 29.56% to 22.36%, retaining 99.04% of control progress. However, the starting controller fails on only 16.88% of the same episodes. The result demonstrates partial mitigation of optimization-induced harm, not a safe controller or an improvement over the starting policy.

CCS CONCEPTS

• **Computing methodologies** → **Bio-inspired approaches**; *Reinforcement learning*.

KEYWORDS

neuroevolution, world models, model exploitation, safety evaluation

1 INTRODUCTION AND RELATED WORK

Learned world models make repeated policy evaluation inexpensive, but search optimizes the learned evaluator rather than the physical system. Errors that appear modest on ordinary trajectories can become consequential when selection repeatedly rewards optimistic predictions. For safety, a policy may appear to avoid failure while violating the constraint when executed.

This problem is not new. Ha and Schmidhuber evolved controllers in learned dreams and used stochasticity to improve transfer [1]. Recent action-sequence audits report that larger proposal pools can worsen selection under latent prediction error [3]. Safety-aware model-based planning also predates this study: Guided Safe Shooting uses MAP-Elites planners [6], while SafeDreamer combines world-model planning and Lagrangian methods [2]. We do not claim that noisy imagination, safe evolutionary planning, or return penalties are new individually.

Our contribution is a controlled empirical audit of *evolved policy* selection in a frozen visual world model. We measure a predeclared change in real-minus-imagined failure as search pressure increases, test a four-arm noise/penalty combination, and separately check actual failure and useful progress. The final comparison uses fresh

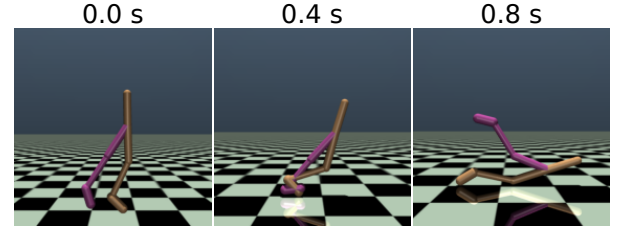


Figure 1: An illustrative control-policy branch, rendered from saved simulator states. The model predicts no health violation, but dense simulator checks first detect one at step 62 (0.496 s). This is the first such final-bank example for the first search seed at high pressure, selected for illustration rather than as a representative effect estimate.

source episodes and freezes every selected policy before inspecting its simulator outcomes. The starting controller remains an explicit reference: reducing the damage caused by search is distinct from improving that controller.

2 POLICY SEARCH AND SAFETY MEASUREMENT

Environment and frozen assets. We use the Walker2d simulator underlying SafetyWalker2dVelocity-v1, with a *health* constraint: torso height must remain strictly between 0.8 and 2.0 m, and pitch between -1 and 1 rad. This differs from the benchmark’s velocity-cost constraint. Actual failure means any violation at one of the 100 simulator steps in a 0.8 s branch. Termination is disabled for branch measurement, retaining post-failure states and progress. “Real” below means execution in MuJoCo, not on physical hardware.

A previously trained Walker2d LeWorldModel (LeWM) [4], its 192-dimensional encoder, physical readouts, and six-member visual-history base controller remain frozen. We evolve a linear residual with 1,020 parameters (60×17) on a fixed 16-dimensional projection plus bias. Each visual update produces ten sequential six-motor actions; actions are clipped to $[-1, 1]$. Visual feedback runs at 12.5 Hz and actions at 125 Hz. The base gain is fixed at one. The residual starts at zero, with RMS-preconditioned coordinates and initial search scale chosen for an expected unclipped action RMS of 0.05.

Noise and scoring. At each predicted block, we perturb the latent before readout and subsequent policy feedback:

$$z_{t+1} = M(z_{t-2:t}, a_{t-2:t}) + k \sigma \odot \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, I). \quad (1)$$

The fixed per-coordinate scale σ is the sample standard deviation of one-step teacher-forced residuals on earlier development data. It is not a state-dependent uncertainty estimate or a calibrated safety bound. We compare $k \in \{0, 1\}$. The noisy latent enters later history windows as well as the policy and health readout.

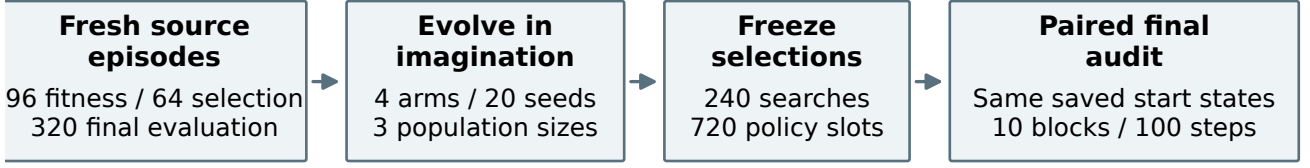


Figure 2: Locked study design. Source episodes are disjoint across roles, with one root per episode. All selections precede final evaluation. Policies close the loop on predicted latents in imagination and on newly rendered observations in MuJoCo; paired starts do not imply identical action tapes.

Let v indicate any imagined health violation and u denote imagined forward progress strictly before the first violating block, or the full horizon if safe. The two fitness rules are

$$F_0 = \mathbb{E}[u], \quad F_P = \mathbb{E}[u + v(V_{\min} - V_{\max})]. \quad (2)$$

Both rules truncate the scored task return identically; the control simply has no additional violation penalty. $V_{\min}$ and $V_{\max}$ are cumulative extrema of observed task returns from that search’s fitness batches, updated once per generation before ranking. This is a ROSARL-style adaptation of the return-range penalty [5], not the original algorithm or a transfer of its safety guarantees. Rollout computation itself continues after violation.

Evolution and selection. Separable CMA-ES [7] searches with populations 16, 64, and 256 for 16 generations. Each generation uses 32 roots sampled without replacement from 96 fitness episodes, shared across arms for a paired seed/generation. Noisy fitness uses four samples per root. One nominee per generation is rescored on 64 fixed selection roots, using 16 samples for $k = 1$ and the current fitness-only penalty bounds. At generations 1, 4, and 16, the best of the baseline and all nominees so far is selected. Zero-noise queries use one sample. Candidate noise streams are shared within each role, while fitness, selection, and final evaluation use separate streams.

3 EXPERIMENTAL PROTOCOL

The design in Figure 2 uses twenty paired search seeds, giving $20 \times 4 \times 3 = 240$ searches and 720 selected policy slots. The model, controller, noise scale, sample size, endpoints, and analysis were locked before collecting 480 fresh source episodes. A four-seed development pilot informed a timing and precision review; the final 20-seed design fit a predeclared 12-hour cap. Earlier controller-readiness and joint transfer screens had failed, and the continuation did not relabel them as passed.

Episodes come from a fixed competent-actor pool with action-noise scales 0, 0.05, and 0.10. Collection stops at the first health violation or 1,000 steps. For each sufficiently long episode, one root is sampled with a three-frame history and room for a 100-step recorded continuation. This conditions the source distribution on survival of the collection policy around the root; it is not an arbitrary-start safety evaluation. The 96/64/320 fitness, selection, and final roles have disjoint episode seeds. The final 320 episodes are

reused across policies for pairing, not treated as 6,400 independent observations per method.

Every selected policy is executed from the same final snapshots used in its imagined audit. It is audited at both $k = 0$ and $k = 1$, with 32 samples per root for the latter. The declared primary gap uses each method’s own noise setting:

$$D_{m,p} = \bar{v}_{m,p}^{\text{real}} - \bar{v}_{m,p}^{\text{imag}}, \quad A_m = D_{m,H} - D_{m,L}, \quad (3)$$

where $L = (16, 1)$ and $H = (256, 16)$ specify population and generation. The two co-primary contrasts are A_C for control and $A_{NP} - A_C$ for noise plus penalty versus control. Gap reduction alone could reflect increased imagined failure, so actual failure and progress are assessed separately.

We use 10,000 paired bootstrap replicates, independently resampling search seeds and source episodes with replacement, preserving method/pressure pairing. The two primary intervals have 97.5% coverage each (Bonferroni family coverage 95%). Other intervals are descriptive 95% intervals. Practical criteria require a negative actual-failure difference and a lower progress-ratio bound above 0.90. Inference is conditional on the frozen assets and shared fitness/selection banks.

4 RESULTS

Both declared primary comparisons are supported. The control gap increases by 10.39 percentage points (pp), with adjusted 97.5% interval [6.70, 14.25]. Noise plus penalty reduces this amplification by 7.22 pp, with contrast interval [−12.18, −2.49] (Figure 4). These are contrasts between low and high pressure, not the high-pressure gap itself. Figure 3 shows all grid cells.

Actual failures fall relative to the evolved control. Figure 1 illustrates a missed violation. At high pressure, the combined method fails on 22.36% of paired evaluations versus 29.56% for control (Table 1). The paired difference is −7.20 pp, descriptive 95% interval [−10.98, −3.70]. Mean full-horizon forward progress is 2.146 versus 2.167 m: the ratio is 0.9904, interval [0.9824, 0.9978], exceeding the 0.90 retention threshold. Both practical criteria pass, with a small measured progress reduction of about 0.96%.

The four arms separate noise and penalty descriptively. At high pressure, adding the penalty without noise changes actual failure by −2.09 pp ([−5.39, 0.59]); with noise the change is −3.30 pp

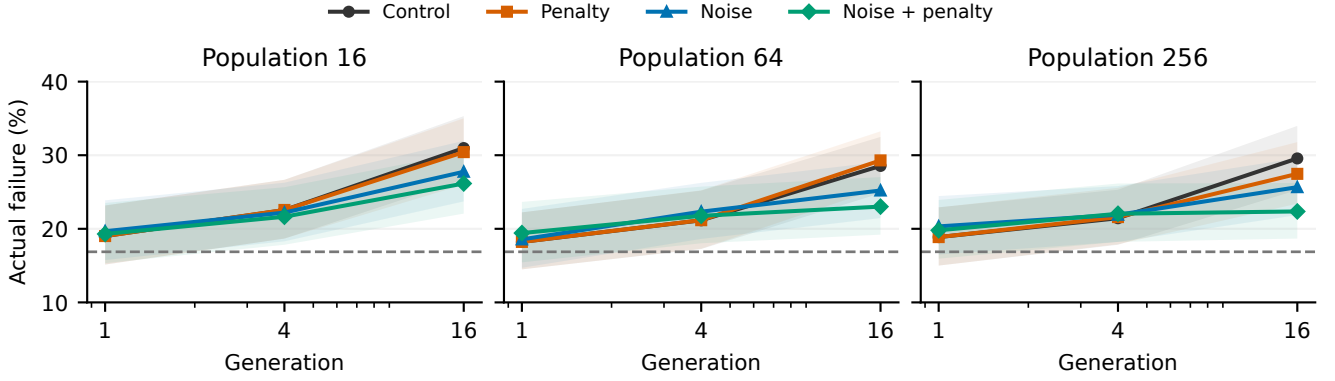


Figure 3: Actual simulator failure across the complete population/generation grid. Shading shows descriptive pointwise 95% crossed-bootstrap intervals. The dashed line is the frozen baseline’s 16.88% failure rate. Larger search budgets generally worsen control safety; the primary test compares the declared endpoints rather than asserting monotonicity. Curves share the same 320 final source episodes.

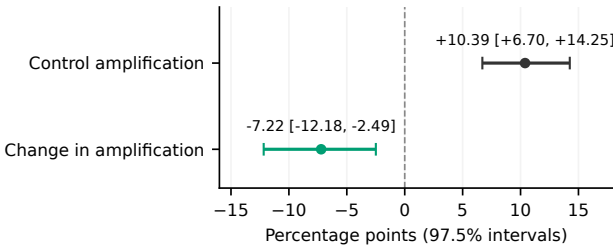


Figure 4: The two co-primary contrasts. Positive control amplification means a larger real-minus-imagined gap. Negative change in amplification means that noise plus penalty reduces this increase relative to control.

Table 1: High pressure: population 256, generation 16. Imagined failure uses each arm’s own noise; the baseline row uses $k = 0$. Evolved rates average 20 seeds on 320 shared episodes. Full-horizon progress includes post-failure motion.

Method	Actual (%)	Imagined (%)	Progress (m)
Control	29.56	0.47	2.167
Penalty	27.47	0.47	2.162
Noise	25.66	5.09	2.155
Noise + penalty	22.36	4.95	2.146
Frozen baseline	16.88	0.31	2.141

($[-6.16, -0.48]$). These secondary intervals are not multiplicity-adjusted and do not establish an interaction. The noisy search also uses more model samples, so a noise-specific effect at equal model-compute cost is not isolated.

The starting controller remains safer. The baseline fails on 54/320 episodes (16.88%). The combined method increases failure relative to it by 5.48 pp ($[2.56, 8.44]$). The result therefore supports *partial*

Table 2: Main-study compute. A predictor row is one latent transition per sample. Final audits include selected policies, the baseline, and reference action tapes.

Phase	Simulator steps	Predictor rows
Source collection	463,921	0
Fitness	0	344,064,000
Selection	0	22,195,200
Final audits	23,104,000	78,144,000
Total	23,567,921	444,403,200

mitigation of harm from optimization, not improvement over the starting controller. Imagined failure is still much lower than actual failure: 4.95% versus 22.36% for the combined method. These imagined probabilities are not calibrated risks, and the primary gap is not conditional false-safe acceptance among accepted policies.

5 LIMITATIONS AND CONCLUSION

This study covers one environment, one frozen model/controller combination, and short branches on a survival-conditioned source distribution. Development choices, including $k = 1$, were informed by earlier observations, although final episodes and analysis endpoints were separated prospectively. Model seeds, fitness-bank variation, full episodes, other environments, and a real-fitness search control are not tested. Baseline health failures and the failed earlier transfer screens preclude a safe-controller claim.

Table 2 separates collection, search, and audit costs. The main run completed in 6.819 hours with zero gradient updates. The preserved evolution-stage ledger totals 26,838,972 simulator steps, with 3,600,058 additional shared original collection steps. Historical actor-training and other upstream costs remain unquantified; these are not lifetime totals. An independent audit checked 118 frozen source files and 11,322 numeric query archives and reproduced the

analysis from saved outputs. The exact main bundle is privately archived; it is not yet an anonymous public replication package.

Evolutionary pressure measurably amplifies a safety-relevant model error. Residual-scaled latent noise and a return-range penalty reduce that amplification and actual failures against the evolved control, but do not undo all damage from search. Independent simulator evaluation and comparison with the starting policy are essential to interpreting the apparent improvement.

REFERENCES

- [1] David Ha and Jürgen Schmidhuber. 2018. Recurrent World Models Facilitate Policy Evolution. In *Advances in Neural Information Processing Systems*, Vol. 31. https://papers.nips.cc/paper_files/paper/2018/hash/2de5d16682c3c35007e4e92982f1a2ba-Abstract.html
- [2] Weidong Huang, Jiaming Ji, Chunhe Xia, Borong Zhang, and Yaodong Yang. 2024. SafeDreamer: Safe Reinforcement Learning with World Models. In *International*

- Conference on Learning Representations*. https://proceedings.iclr.cc/paper_files/paper/2024/hash/ece182f93af26c64187ba3f7dfd4309a-Abstract-Conference.html
- [3] Liangyu Li, Qingwen Liu, Mingqing Liu, and Wen Fang. 2026. Action from Adjacent Set in Physical Space Outperforms the Best Prediction in World Models. *arXiv preprint arXiv:2607.23602* (2026). <https://doi.org/10.48550/arXiv.2607.23602>
- [4] Lucas Maes, Quentin Le Lidec, Damien Scieur, Yann LeCun, and Randall Balestriero. 2026. LeWorldModel: Stable End-to-End Joint-Embedding Predictive Architecture from Pixels. *arXiv preprint arXiv:2603.19312* (2026). <https://doi.org/10.48550/arXiv.2603.19312>
- [5] Geraud Nangue Tasse, Mark Nemecek, Tamlin Love, Steven James, and Benjamin Rosman. 2026. An Unreasonably Simple Approach to Safe RL. *Reinforcement Learning Journal* 7 (2026). <https://rlj.cs.umass.edu/2026/papers/Paper96.html>
- [6] Giuseppe Paolo, Jonas Gonzalez-Billandon, Albert Thomas, and Balázs Kégl. 2022. Guided Safe Shooting: Model Based Reinforcement Learning with Safety Constraints. *arXiv preprint arXiv:2206.09743* (2022). <https://doi.org/10.48550/arXiv.2206.09743>
- [7] Raymond Ros and Nikolaus Hansen. 2008. A Simple Modification in CMA-ES Achieving Linear Time and Space Complexity. In *Parallel Problem Solving from Nature – PPSN X*. 296–305. https://doi.org/10.1007/978-3-540-87700-4_30